

Guide: using the bounded transfer cutoff

Collatz Lean research project

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The gap this repairs

Knowing the transfer statements $A(v)$ for $v < u$ does not automatically give convergence of every integer below u . The odd-run proof can request a larger transfer parameter: seed 511 invokes parameter 546. The new theorem supplies an explicit threshold that the smaller-transfer hypotheses really do justify.

How to apply the rule

Given all $A(v)$ for $v < u$, choose b with $3^b \leq 36u$. It is then legitimate to use convergence of every positive integer below 2^b . If the target $27u + 20$ reaches that interval, the transfer at u follows.

The cutoff is obtained entirely with integers. The search multiplies by three until the next power would exceed $36u$. No rounded logarithms are used. An unsuccessful bounded search remains a depth limit, not a counterexample.

Formal entry points

In `lean/Collatz/BoundedTransfer.lean`:

- `odd_run_size_bound` and `odd_run_transfer_size` justify the size accounting.
- `reachesOne_below_pow_two` proves convergence below 2^b from transfer instances satisfying $36v + 27 \leq 3^b$.
- `reachesOne_of_smaller_transfers` connects this cutoff to the strong-induction hypotheses.
- `AffineTransfer.of_dyadic_descent` closes a transfer from a descent into the justified interval.

For Mersenne exits, compose the target identity from `MersenneExit.lean` with this last rule. The finite search checks the exit state, so its recorded clock is additional to the growing prefix.

Reproduce

From the repository root, with the pinned toolchain on PATH:

```
python3 analysis/mersenne_safe_descent.py
python3 -m unittest discover -s analysis \
  -p 'test_mersenne_safe_descent.py'
python3 analysis/audit_theorems.py --build
```

From `lean/`, run `lake build Collatz.BoundedTransfer`. The saved finite witnesses are in `analysis/mersenne_safe_descent_results.json`.

What remains

All 513 tested Mersenne exits descend to the justified cutoff, but this does not establish the rule for every exponent or every parameter. The next mathematical requirement is a general descent argument, or a covering combination of descent and transfer rules. Smaller-transfer premises must remain explicit until that induction is complete.